Engineering Workshop Practice

Complete practical handbook covering industrial safety, carpentry, fitting, welding, sheet metal work, electrical wiring, and basic electronics with hands-on approach.

Credits	1.5	Prerequisite	Engineering Graphics (1003)
Contact Hours	45	Course Type	Practicum / Lab
Self-Learning	45 Hours	Programme	Polytechnic Diploma Programmes
CIE Marks	60	Exam Scheme	CIE + ESE Practical
ESE Marks	40		
Total Marks	100		

Official Source Declaration

This handbook is derived from the **2009A Engineering Workshop Practice** syllabus (Kerala Diploma Revision 2026). The content aligns with the official PDF document provided by the board.

Verified Inconsistency / Interpretation:
The official PDF exhibits an inconsistency between the Detailed Syllabus section and the Student Activities table. The Detailed Syllabus (Page 2-3) correctly labels the Sheet Metal section as **Module V**. However, the "Student Activities for Self-Learning" table on Page 6 incorrectly titles this section as "Module IV" (likely a typographical error by the publishers).

Resolution: This handbook follows the logical ordering of the Detailed Syllabus (Modules I through VI) while incorporating the self-learning activities into their respective modules. The sheet metal section is correctly presented as **Module V** to match the course objectives and instructional content.

Course Dashboard

45

Lab Contact Hours

1.5

Credits

60+40

CIE + ESE

Teaching and Assessment Scheme

Teaching Scheme / Week	Self-Learning / Week	Total Hours / Sem	Credits	CIE	ESE	Total
P: 3	1.5	45	1.5	60	40	100

Pre-requisites

- Trigonometry and basic mathematical principles (Secondary School).
- Principles of electrical conductivity and basics of electronics (Secondary School).
- Geometric constructions and projections (1003 - Engineering Graphics, Sem I).

Key Competencies

- Industrial safety and fire extinguisher operation.
- Wood working, fitting, welding, and sheet metal fabrication.
- Electrical wiring, motor connection, and soldering techniques.

How to Use This Handbook

📖 Understand

Read the detailed explanations for each topic, including safety precautions and theoretical concepts.

📐 Visualise

Study the SVG diagrams, compare tables, and use the interactive calculators to see principles in action.

🛠️ Apply

Follow the step-by-step procedures for making joints, wiring, and soldering.
Practice with the self-learning activities.

✅ Revise

Use the Q&A, model paper, revision cards, and formula bank to prepare for CIE and ESE exams.

Course Outcomes & Objectives

Course Objectives

- To familiarize safety precautions practiced in the workplace.
- To prepare drawings of models for fabrication.
- To identify and practice various measuring, marking, holding, striking, cutting tools and precision instruments.
- To practice electrical wiring and soldering.
- To familiarize with basic electronic components.
- To operate machines, power tools, and equipment safely.

Course Outcomes (COs)

Bloom's Taxonomy Mapping

CO	Course Outcome	Bloom's Level
CO1	Identify the importance of industrial safety and hands-on fire extinguisher practice and the tools and devices required to make carpentry joints and familiarise various wood alternative materials (MDF, Multi wood, plywood, etc.)	Apply
CO2	Make use of various tools, machines, instruments and power tools used in the fitting shop to make fitting joints.	Apply
CO3	Make use of various tools, machines, instruments and power tools used in the welding shop to make welding joints.	Apply
CO4	Make use of various tools and measuring instruments to fabricate sheet metal models.	Apply
CO5	Make use of various tools and accessories to practice electrical wiring, motor connection and soldering of basic electronic circuits.	Apply

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO2	3	-	-	-	-	-	-	-	-	-	-
Mapping Key: 1 - Low; 2 - Medium; 3 - High; - No Correlation											

Curriculum Coverage Map

Module to Topic Mapping

Module	Key Topics	CO	Hours	Handbook Section
I	Industrial Safety, Fire Extinguishers	CO1	3	Module I
II	Carpentry Tools, Materials, Open Halved Joint	CO1	12	Module II
III	Fitting Tools, Precision Instruments, Fitting Joints	CO2	12	Module III
IV	Welding Safety, Arc Welding, Butt Joints, Pipe Fabrication	CO3	9	Module IV
V	Sheet Metal Tools, Operations, Development of Tray/Funnel	CO4	12	Module V
VI	Electrical Wiring, Motor Connection, Electronics, Soldering	CO5	12	Module VI

Module Roadmap

🛡️ Foundation (Mod I)

Start with **Industrial Safety**. Understand fire classes, extinguisher types, and personal safety protocols. This knowledge is applied throughout the workshop.

🪚 Material Shaping (Mod II, III, IV)

Progress through **Carpentry** (wood), **Fitting** (metal), and **Welding** (joining metals). Learn to operate power tools and measure with precision instruments.

📐 Fabrication & Finishing (Mod V)

Develop skills in **Sheet Metal** work. Apply geometric developments to create practical models like trays, funnels, and dustpans.

⚡ Electrical & Electronics (Mod VI)

Conclude with **Electrical wiring** (staircase, motor starter) and **Basic Electronics** (rectifiers, soldering). Connect theoretical circuits to real hardware.

Module I: Industrial Safety

CO1 | 3 Hours | Understand

🎯 Learning Goals

Understand importance of safety, identify hazards, and master fire extinguisher operation.

📌 Key Facts

Safety is a 3-pronged approach: Personal, Equipment, and Environmental safety.

🛠️ Engineering Habit

Always identify the nearest fire extinguisher and emergency exit before starting any work in a new environment.

1. Importance of Safety in Industry & Hazardous Situations

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Why Safety Matters: In an industrial setting, safety is not just a rule—it is a fundamental prerequisite for efficient operations. Accidents lead to loss of life, injuries, damage to equipment, and financial losses. A safe working environment boosts employee morale and productivity.
Hazardous Situations: These are conditions that can cause harm. Common industrial hazards include:

- **Physical Hazards:** Moving machinery parts, sharp edges, noise, vibration, and falling objects.
- **Chemical Hazards:** Exposure to toxic fumes, acids, solvents, and flammable liquids.
- **Electrical Hazards:** Exposed live wires, faulty insulation, and overloaded circuits.
- **Ergonomic Hazards:** Repetitive motions, improper lifting techniques, and poor workstation design.

- **Psychological Hazards:** Stress, fatigue, and overwork leading to poor concentration.

> **Malayalam support:** [illegible]

2. Personal Safety & Precautions

Personal Protective Equipment (PPE): PPE is the last line of defense against injuries. It includes:

- **Head:** Safety helmets to protect against falling objects and impacts.
- **Eyes & Face:** Safety goggles and face shields to protect against flying debris, chemical splashes, and radiation (e.g., welding arc).
- **Ears:** Earplugs or earmuffs in high-noise areas (e.g., grinding, forging).
- **Hands:** Leather gloves for rough materials, rubber gloves for electrical work, and heat-resistant gloves for welding.
- **Feet:** Steel-toed safety shoes to prevent crushing and puncture injuries.
- **Body:** Aprons and coveralls to protect from sparks, chemicals, and dust.

General Precautions:

- Always read and follow safety signs and warning labels.
- Do not operate machinery without proper training and authorization.
- Keep work areas clean and free of clutter (5S methodology).
- Never wear loose clothing, jewelry, or tie long hair near moving machinery.
- Report any unsafe conditions or equipment malfunctions immediately to the supervisor.

> **Safety First:** "It is better to be safe than sorry." A single moment of carelessness can lead to a lifetime of disability.

3. Types of Fire, Classes of Fire, and Methods of Extinguishing

Understanding fire classes is crucial for choosing the correct extinguisher.

Classes of Fire and Extinguishing Agents

Class	Material / Source	Suitable Extinguisher	Method
A	Ordinary combustibles (Wood, Paper, Cloth, Plastics)	Water, Foam, Dry Powder	Cooling (Water) or Smothering (Foam/Powder)

Class	Material / Source	Suitable Extinguisher	Method
B	Flammable Liquids (Petrol, Oil, Paint, Solvents)	Foam, Dry Powder, CO_2	Smothering / Starving (Removing Oxygen)
C	Flammable Gases (LPG, CNG, Acetylene)	Dry Powder, CO_2	Smothering
D	Combustible Metals (Magnesium, Titanium, Sodium)	Special Dry Powder (Class D)	Smothering (Requires special agent)
K	Cooking Oils & Fats (Kitchen fires)	Wet Chemical (Potassium Acetate)	Cooling and forming a soapy foam barrier

Methods of Fire Extinguishing:

- **Cooling:** Reducing the temperature below the ignition point (using water).
- **Smothering:** Removing oxygen from the fire triangle (using Foam, CO_2, or Dry Powder).
- **Starvation:** Removing the fuel source.
- **Chemical Inhibition:** Interrupting the chemical chain reaction (using Halon or Dry Chemical).

> **Malayalam support:** [illegible]

4. Hands-on Practice on Fire Extinguishers

Procedure to Operate a Fire Extinguisher (P.A.S.S. Technique):

1. **P - Pull** the safety pin located on the handle. This breaks the tamper seal.
2. **A - Aim** the nozzle or hose at the *base* of the fire (not the flames).
3. **S - Squeeze** the lever handle to release the extinguishing agent.
4. **S - Sweep** from side to side, moving the nozzle back and forth across the base of the fire until it is extinguished.

Key Safety Guidelines:

- Always position yourself with an exit behind you.
- Maintain a safe distance (typically 8–10 feet).
- Do not inhale the agent; use it in a well-ventilated area.

- If the fire is too large or spreading rapidly, evacuate immediately and call emergency services.

> 💡 **Practical Tip:** During hands-on practice, simulate the procedure using a dummy extinguisher or a water-based unit to build muscle memory. Remember the acronym **P.A.S.S.**

> **Module I Summary:** Safety is the foundation of all workshop activities. Knowing fire classes, proper PPE usage, and how to correctly operate a fire extinguisher using the P.A.S.S. technique ensures a secure and productive working environment.

Module II: Carpentry & Wood Working

CO1 | 12 Hours | Apply

🎯 Learning Goals

Identify woodworking tools, understand alternative wood materials, and fabricate a simple open halved joint.

📌 Key Facts

Carpentry is one of the oldest crafts. Modern alternatives like MDF and WPC offer durability and ease of use.

🛠️ Engineering Habit

Always measure twice and cut once. Precision is key to achieving a tight-fitting joint.

1. Demonstration of Wood Working Tools & Power Tools

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Common Carpentry Hand Tools and Their Uses

Tool Category	Tool Name	Purpose / Function

Tool Category	Tool Name	Purpose / Function
Marking	Marking Gauge	To mark lines parallel to the edge of the wood.
	Try Square	To check and mark 90-degree angles.
	Scriber / Pencil	To scratch or draw reference lines.
Planning	Jack Plane	To smooth and flatten the surface of the wood by removing thin shavings.
Sawing	Rip Saw	To cut wood along the grain (ripping).
	Cross-cut Saw	To cut wood across the grain.
	Tenon Saw	To make fine, precise cuts for jointing.
Chiseling	Chisel (Firmer / Paring)	To carve out material, create mortises, and clean up joint surfaces.
Holding	Carpenter's Vice	To firmly hold the workpiece during operations.
Striking	Mallet / Hammer	To drive chisels and assemble joints.
Drilling	Hand Drill / Brace and Bit	To make holes in the wood.

Common Power Tools in Carpentry

Tool Name	Function
Electric Hand Circular Saw	Quickly cuts large sheets of wood and plank with high speed.
Electric Hand Planer	Removes material and smooths surfaces much faster than a manual plane.
Electric Drill	Drills precise holes quickly.

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2. Familiarisation of Various Alternate Materials of Wood

Traditional solid wood is becoming scarce and expensive. Modern workshops heavily utilize engineered wood products.

Alternate Wood Materials

Material	Composition	Properties	Uses
MDF (Medium Density Fibreboard)	Wood fibres mixed with resin and compressed under heat and pressure.	Uniform density, smooth surface, easy to machine, but not waterproof.	Furniture, interior decoration, cabinet making.
WPC (Wood Plastic Composite)	Recycled plastic and wood dust/ fibres.	High durability, moisture and rot-resistant, eco-friendly.	Outdoor furniture, decking, fencing, and flooring.
Plywood	Thin layers of wood veneer glued together with alternating grain directions.	High strength, resists warping and splitting, waterproof grades available (Marine ply).	Construction, shipbuilding, heavy-duty furniture, and formwork.
Blockboard	Core of softwood strips sandwiched between two layers of veneer.	Lightweight, strong, and cheaper than plywood.	Interior doors, shelves, and low-cost furniture.

3. Demonstration of Wood Working Processes

Marking: The process of transferring measurements from a drawing to the wood surface. Tools used: rule, pencil, marking gauge, and try square. Accuracy in marking prevents material wastage.
Planning: Removing a thin layer of wood to create a flat, smooth surface. Always plane along the grain to avoid tear-out. The jack plane is the most common tool used.
Cutting (Sawing): Using a saw to divide the wood along desired lines. Rip saws cut with the grain, while crosscut saws cut across the grain. Tenon saws are used for fine joinery.
Chiseling: Using a chisel and mallet to remove waste material from a joint (e.g., a mortise or halving joint).
Grooving: Cutting a groove or channel along the length of a piece of wood using a router or a grooving plane.

4. Construct a Simple Joint: Open Halved Joint

The **Open Halved Joint** is one of the simplest and strongest corner joints. It is made by removing half the thickness of each piece so they interlock perfectly.

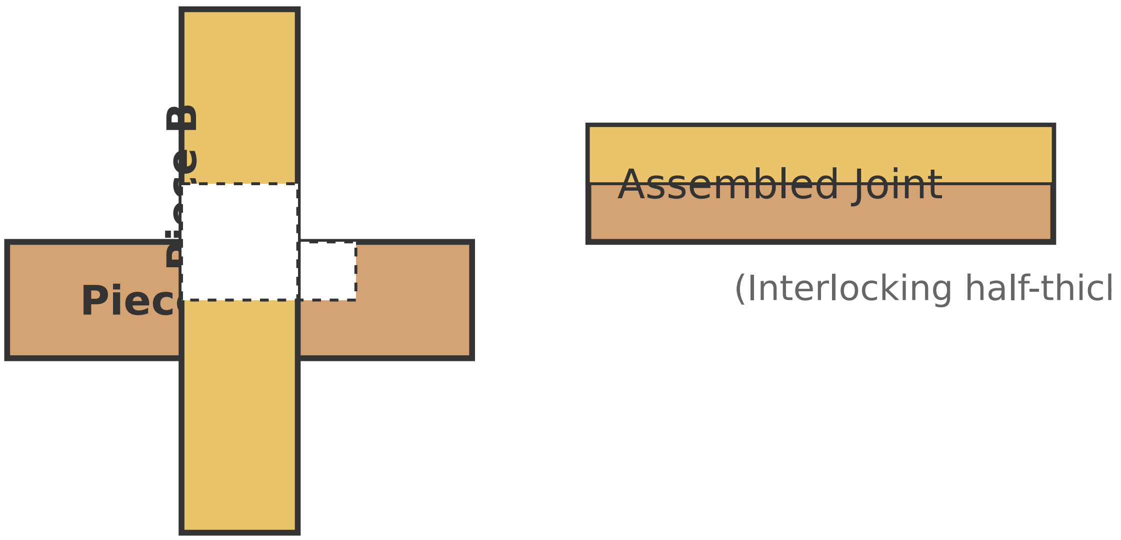

Figure: Open Halved Joint (Exploded and Assembled views)

Procedure to Make an Open Halved Joint:

1. **Marking:** Using a try square and marking gauge, mark the centerline of the joint on both wooden reapers. Mark the half-thickness on the edge.
2. **Sawing:** Use a tenon saw to cut along the marked lines down to the half-thickness depth.
3. **Chiseling:** Remove the waste wood using a mallet and chisel. Chisel from both sides to avoid splintering the edge.
4. **Fitting & Sanding:** Test the fit. The pieces should interlock snugly without gaps. Sand the surfaces for a smooth finish.
5. **Assembly:** Apply wood glue to the joint surfaces and press them together. Clamp until the glue sets.

💡 **Practical Tip:** Always test the fit before applying glue. A tight fit without glue is better than a loose fit with glue.

Module II Summary: Carpentry involves precise measurement, skillful use of hand and power tools, and knowledge of material properties. The ability to create a clean, well-fitting open halved joint demonstrates mastery of basic marking, sawing, and chiseling operations.

Module III: Fitting Shop

CO2 | 12 Hours | Apply

🎯 Learning Goals

Identify fitting tools, perform measuring and cutting operations, and make a fitting joint using precision instruments.

📌 Key Facts

Fitting is the process of removing material to achieve a desired shape or a perfect match between two mating parts.

🛠️ Engineering Habit

Always use the appropriate file for the job. Using a coarse file on a finish surface will ruin the precision.

1. Demonstration of Tools, Machines, Instruments & Power Tools in Fitting Shop

Fitting Shop Tools and Equipment

Category	Tool / Machine	Function
Holding	Bench Vice	To securely hold the workpiece for filing, sawing, and chipping.
Marking & Measuring	Surface Plate, Try Square, Scriber, Steel Rule	To provide a flat reference surface and accurately mark workpieces.
Cutting	Hacksaw	To cut metal bars, pipes, and rods.
Filing	Files (Flat, Round, Square, Triangular, Half-round)	To remove material, smooth edges, and shape the workpiece.
Drilling	Bench Drilling Machine	To drill accurate holes in metal components.
Grinding	Bench / Pedestal Grinder	To sharpen tools and remove burrs from workpieces.
Power Tools	Hand Drilling Machine, Angle Grinder	For portable drilling and cutting/grinding tasks.

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2. Demonstration of Fitting Operations

- **Measuring:** Determining the dimensions of the workpiece using a steel rule or vernier caliper.
- **Marking:** Using a scriber, try square, and surface gauge to transfer the measurement lines onto the metal surface.
- **Cutting:** Using a hacksaw to cut the metal bar to the required length.
- **Filing:** The process of using a file to remove material. There are different strokes and file types for different purposes (roughing, finishing).
- **Drilling:** Creating holes using a drilling machine and a drill bit.

Note on Hacksaw Blade Selection:
- **Coarse blades (14 TPI):** For soft metals like Aluminum and Brass.
- **Medium blades (18-24 TPI):** For general purpose mild steel.
- **Fine blades (32 TPI):** For thin materials like tubing and sheet metal.

3. Construct a Simple Fitting Joint & Use of Precision Measuring Instruments

Fitting Joint: A typical exercise is the **Square Gap Fitting** or an **L-Fitting**. This involves creating a precise, right-angled cut and fitting two pieces together without any gap.

Procedure for Making a Simple Square Fit:

1. **Marking:** Measure and mark the required length on the mild steel bar (e.g., 50mm × 25mm). Use a try square to ensure a right angle.
2. **Cutting:** Use a hacksaw to cut the bar to length. Leave a small margin (1-2mm) for filing.
3. **Filing:** Clamp the workpiece in the bench vice. Use a flat file to remove the remaining material and achieve the exact dimension. Regularly check with a Vernier Caliper.
4. **Checking Squareness:** Use a try square to ensure the edges are perfectly perpendicular.

5. **Assembly:** When two such accurately machined pieces are joined, they form a perfect fitting joint.

Precision Measuring Instruments

Vernier Caliper vs. Micrometer

Instrument	Principle	Least Count (Typical)	Use
Vernier Caliper	Measures the difference between one main scale division and one vernier scale division. L.C. = 1 MSD - 1 VSD.	0.02 mm	Measures inside, outside, and depth of objects.
Micrometer Screw Gauge	Uses a precision screw thread. Pitch of the screw determines the accuracy. L.C. = Pitch / Number of Divisions on thimble.	0.01 mm	Measures very small dimensions (e.g., thickness of a sheet).

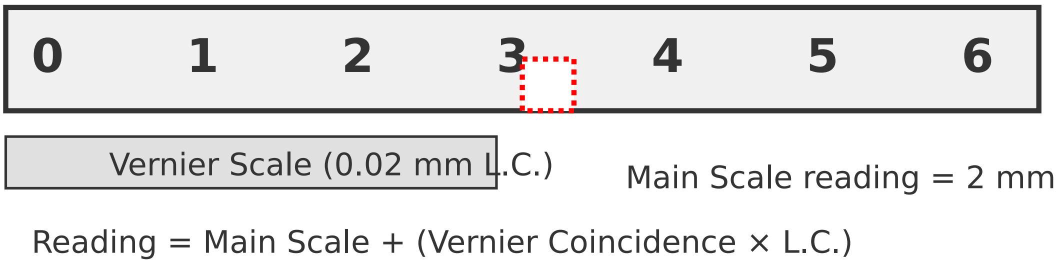

Figure: Principle of Vernier Caliper Reading

Worked Example: Vernier Caliper Reading
Given: 1 MSD = 1 mm, 10 divisions on Vernier scale. L.C. = 1/10 = 0.1 mm.
Observation: Main scale reads 3.2 mm. The 5th division of the Vernier scale coincides with a main scale division.
Calculation: Total Reading = 3.2 mm + (5 × 0.1 mm) = 3.2 + 0.5 = **3.7 mm**.

Module III Summary: Fitting is a fundamental metalworking skill. It requires proficiency in holding, cutting, filing, and measuring. Precision instruments like Vernier Calipers and Micrometers are essential for achieving accurate, high-quality fittings.

Module IV: Welding Shop

CO3 | 9 Hours | Apply

🎯 Learning Goals

Understand welding safety, identify welding tools/PPEs, fabricate simple shapes using arc welding.

📌 Key Facts

Arc welding uses an electric arc to melt and join metals. The electrode provides the filler material.

🛠️ Engineering Habit

Always wear a welding helmet and gloves. The UV rays from the arc can cause severe eye damage (flash burn) and skin burns.

1. Welding Safety Precautions, Tools, Equipment, and PPEs

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Critical Welding Safety Precautions:

- Never look at the arc with the naked eye. Always use a **welding helmet** with the correct shade of filter glass (Shade 10–13).
- Wear **leather welding gloves** and an **apron** to protect from sparks and heat.
- Ensure the work area is well-ventilated to prevent inhalation of fumes.
- Keep flammable materials away from the welding station.
- Check the insulation of the welding cables for any cracks or damage.

Essential Welding Equipment and PPEs

Type	Item	Function
Welding Machine	Arc Welding Machine (SMAW/MMA)	Provides the high current (250A) required to create and maintain the electric arc.
Electrode Holder	Stinger	Holds the electrode and conducts the current to it.
Ground Clamp	Earth Clamp	Completes the electrical circuit by connecting the workpiece to the machine.
PPE	Welding Helmet, Gloves, Apron	Protects eyes, hands, and body from arc flash, UV radiation, and sparks.

Type	Item	Function
Cleaning Tools	Chipping Hammer, Wire Brush	Removes slag and splatter from the weld bead.

2. Practice Straight Line Welding and Square Butt Joints

Straight Line Welding (Beading): The technique of laying a weld bead in a straight line without any joint. This is the first exercise for a beginner to learn arc length, travel speed, and consistency.

Procedure for Straight Line Welding:

1. **Setup:** Clean the mild steel plate (remove rust/dirt) and clamp the earth lead to it.
2. **Striking the Arc:** Scratch the electrode against the plate to create a spark, then lift it to the correct arc length (about 3mm).
3. **Travel:** Move the electrode along the seam at a constant speed. Hold the electrode at an angle of 10-20 degrees from the vertical.
4. **Finishing:** Stop the arc by lifting the electrode quickly.
5. **Cleaning:** Use the chipping hammer and wire brush to remove the slag.

Square Butt Joint: This is the simplest joint, where two plates are placed edge-to-edge and welded along the seam.

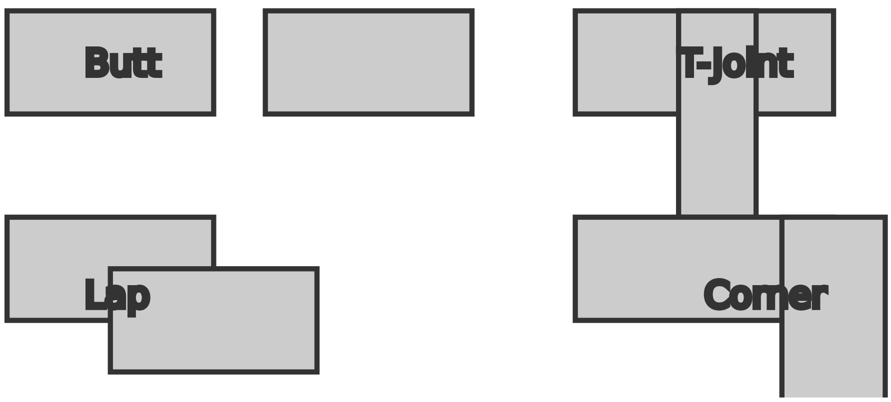

Figure: Common Types of Welding Joints

3. Fabricate Simple Shapes using Square or Round Pipes (Group Work)

Group Fabrication: This exercise involves teamwork and planning. A group of students will cut and weld pipes to form geometric shapes like a **Triangle** or a **Rectangle**.

Procedure for Fabricating a Triangle from Square Pipes:

1. **Marking & Cutting:** Measure three pieces of square pipe (e.g., 30cm each). Mark 45-degree angles at both ends of each piece to form miter joints.
2. **Cutting:** Use a hacksaw or a cut-off saw (chop saw) to cut the pipes at the marked 45-degree angles.
3. **Fitting (Tack Welding):** Arrange the three pieces on a flat surface to form a triangle. Use **tack welds** (small, temporary welds) at the three corners to hold the pieces in place.
4. **Welding:** Weld all the seams completely, ensuring good penetration.
5. **Finishing:** Grind the weld beads smooth using an angle grinder.

💡 **Practical Tip:** Use a square to check the 90-degree corners of the rectangle, and use trigonometry to calculate the lengths for the triangle.

Module IV Summary: Welding is a critical joining process. Mastering the technique requires a balance of safety awareness, proper equipment usage, and steady hand coordination. Fabricating simple shapes from pipes develops teamwork and precision welding skills.

Module V: Sheet Metal Shop

CO4 | 12 Hours | Apply

🎯 Learning Goals

Identify sheet metal tools, perform operations like cutting and bending, and develop models like trays and funnels.

📌 Key Facts

Sheet metal work involves shaping thin metal sheets. Development is the process of unfolding a 3D shape into a 2D flat pattern.

🛠️ Engineering Habit

Always account for the thickness of the material and the bend allowance when marking out the development.

1. Demonstration of Tools and Equipment in Sheet Metal Shop

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Sheet Metal Hand Tools

Tool Name	Function
Hand Shear (Snips)	Used to cut sheet metal along straight or curved lines.
Mallet	A soft-faced hammer (rubber or wooden) used to bend sheet metal without damaging the surface.
Stakes	Metal anvils of various shapes (T-stake, Round stake) used to provide a surface against which the sheet is formed.
Folding Bar	Used to hold the sheet firmly while bending edges.
Dividers / Compass	Used to scribe circles and arcs on the sheet metal.

2. Demonstration of Sheet Metal Operations

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- **Sheet Cutting:** Using hand shears (snips) or a guillotine shearing machine to cut the sheet to the required size.
- **Bending:** Using a mallet and a folding bar to bend the sheet along a straight line.
- **Edging:** Bending the edge of a sheet to create a flange. This increases the stiffness and eliminates sharp edges.
- **Hemming:** Folding the edge of the sheet over itself.
- **Seaming:** Joining two edges of sheet metal together. Common types are the **Lap seam**, **Grooved seam**, and **Single seam**.

3. Development of Geometrical Models (Tray, Box, Funnel)

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Development (Pattern Making): The process of laying out the flat sheet metal pattern that, when folded and formed, will create the desired 3D object.

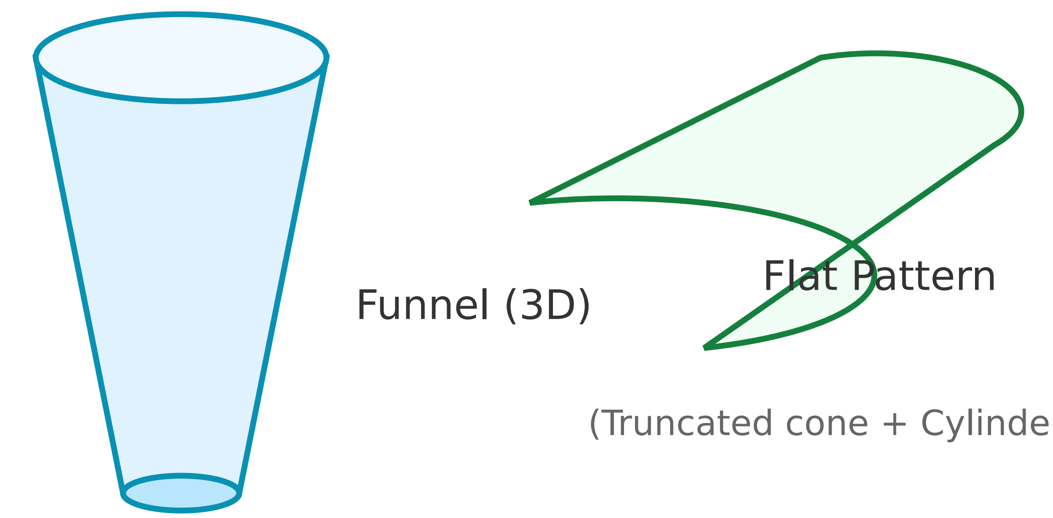

Figure: Sheet Metal Development of a Funnel

Constructing a Sheet Metal Funnel:

1. **Pattern Marking:** Use a compass and steel rule to mark the development of the tapered part (a sector of a ring) and the cylindrical part (a rectangle) on the sheet metal.
2. **Cutting:** Cut out the two flat patterns using snips.
3. **Forming:** Roll the conical part over a round stake to form the taper. Roll the rectangular part over a cylindrical stake.
4. **Joining (Seaming):** Overlap the edges of the conical part and the cylindrical part. Use a grooved seam or solder to join the seams.
5. **Finishing:** Remove sharp edges and hammer the joints flat.

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Module V Summary: Sheet metal work combines geometry and fabrication. Mastery of development techniques allows a craftsman to create complex three-dimensional objects from a simple flat sheet. Tray, box, and funnel are standard projects that teach these fundamental concepts.

Module VI: Electrical Wiring & Electronics

CO5 | 12 Hours | Apply

🎯 Learning Goals

Practice electrical wiring (staircase), connect a motor with DOL starter, identify electronic components, and perform soldering.

📌 Key Facts

Electrical wiring is governed by strict safety codes. DOL starter provides a controlled start for induction motors.

🛠️ Engineering Habit

Always use a multimeter to check for live wires before touching any exposed electrical circuit.

1. Electrical Safety, Wiring Tools, and Materials

Electrical Safety Precautions:

- Always disconnect the power supply before making any changes to an electrical circuit.
- Ensure proper insulation of all wires to prevent short circuits.
- Use fuse or MCB (Miniature Circuit Breaker) to protect circuits from overload.
- Never handle electrical equipment with wet hands.
- Proper earthing is essential to prevent electric shock.

Electrical Wiring Tools and Materials

Type	Item	Purpose
Tools	Screwdriver, Pliers, Wire Stripper, Tester	For connecting wires, stripping insulation, and checking live circuits.
Wiring Materials	PVC Wires (Single core / multi-core), Conduit pipes, Switch boxes	To carry current and protect the wiring.
Components	Switches, Sockets, Bulb holders, Fuses	For controlling and protecting the circuit.

2. Practice on Electrical Wiring (One Lamp – One Switch & Staircase Wiring)

One Lamp Controlled by One Switch: The simplest electric circuit. The switch is connected in the live/phase wire. When the switch is closed, the circuit is complete and the lamp glows.
Staircase Wiring (Two-Way Switch): This allows the lamp to be turned ON and OFF from two different locations (e.g., top and bottom of a staircase).

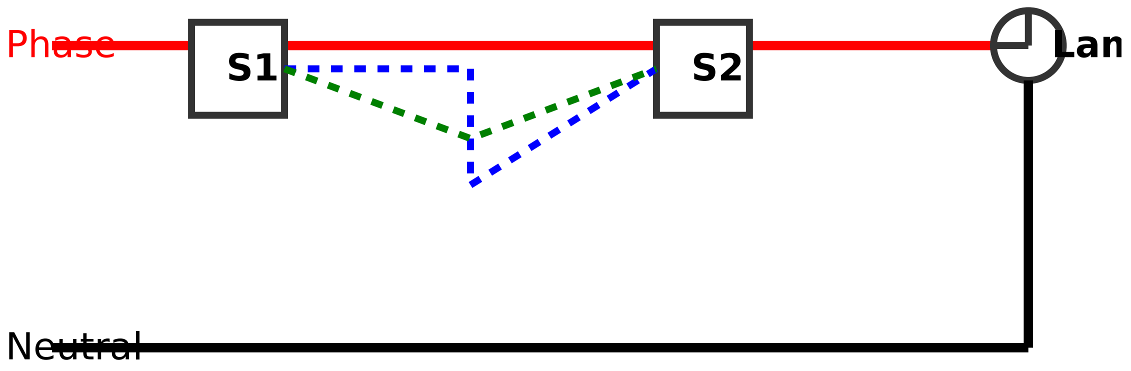

Figure: Staircase Wiring with Two-Way Switches

3. Connection of Single Phase Induction Motor with DOL Starter

DOL (Direct On Line) Starter: It is the simplest and most common method for starting single-phase induction motors up to 5 HP. It provides a full voltage start.
Components: A DOL starter consists of a **Contactor** (to switch the power), **NO/NC Push Buttons** (Start and Stop), and an **Overload Relay** (to protect the motor from overcurrent).

Procedure for Connecting a Motor with DOL Starter:

1. Identify the main supply wires (Phase, Neutral, and Earth).
2. Connect the Phase wire to the input of the MCB, then to the Contactor's L1 terminal.
3. Connect the Neutral wire directly to the motor's Neutral terminal.
4. Connect the "Start" (NO) and "Stop" (NC) push buttons to the Coil circuit of the Contactor.
5. Connect the output of the Contactor (T1, T2, T3) to the motor's terminals.

6. Connect the Overload Relay in series with the motor circuit to sense the current.

> 💡 **Practical Tip:** Always check the motor's voltage rating (230V for single-phase) before connecting. Ensure the starter is properly earthed.

4. Identification of Electronic Components & Soldering Practice

×

Basic Electronic Components

Component	Symbol	Function	Type
Resistor	Zig-zag line	Limits the flow of current.	Passive
Capacitor	Two parallel lines	Stores electrical charge.	Passive
Diode	Triangle with a line	Allows current to flow in one direction only (Rectification).	Active
Transistor	Three terminals (B, C, E)	Acts as a switch or amplifier.	Active
Integrated Circuit (IC)	Rectangle with pins	Performs specific circuit functions (amplification, logic, etc.).	Active

Half Wave and Full Wave Rectifiers:

- **Half Wave:** Uses a single diode to convert AC to pulsating DC. It only conducts during one half of the AC cycle.
- **Full Wave (Bridge):** Uses four diodes in a bridge configuration to convert AC to DC. It conducts during both halves of the AC cycle, making it much more efficient.
- **Filter Circuit:** A capacitor is connected across the output of the rectifier to smooth out the pulsating DC into a steady DC voltage.

Soldering Iron and Accessories

Tool / Material	Purpose
Soldering Iron	Provides heat (usually 25W-60W) to melt the solder.
Solder Wire	An alloy of Tin and Lead (60/40) that joins the components.
Flux	Cleans the surfaces to be joined, ensuring a good solder joint.
PCB (Printed Circuit Board)	A board with copper tracks used to mount and connect electronic components.

Procedure for Soldering:

1. **Tinning:** Heat the soldering iron tip and apply a small amount of solder to coat the tip.
2. **Heating the Joint:** Place the iron tip against the joint of the component lead and the PCB pad for about 2 seconds.
3. **Applying Solder:** Feed the solder wire into the junction. When the metal reaches the melting temperature, the solder will flow smoothly.
4. **Removing Heat:** Remove the iron and the solder wire simultaneously. Let the joint cool naturally.
5. **Inspection:** A good joint should appear shiny and have a volcano-like shape.

Module VI Summary: Electrical and electronic circuits are the backbone of modern industry. Understanding wiring diagrams, motor starters, and component identification is essential for any technician. Soldering is a fundamental skill that must be practiced carefully.

Formula Bank

Ohm's Law

$$V = I \times R$$

Voltage = Current × Resistance

Electrical Power

$$P = V \times I$$

Power (Watts) = Voltage × Current

Energy Consumption

$$E = P \times t$$

Energy (kWh) = Power (kW) × Time (hours)

Vernier Caliper L.C.

$$L.C. = 1\ MSD - 1\ VSD$$

Least Count = 1 Main Scale Div - 1 Vernier Scale Div

Micrometer L.C.

$$L.C. = Pitch / No.\ of\ Thimble\ Divisions$$

Least Count = Screw Pitch / Divisions on Circular Scale

Resistor Color Code

$$R = (A \times B) \times 10^{C} \pm D$$

A=1st Band, B=2nd Band, C=Multiplier, D=Tolerance

Diagram Library for Rapid Revision

🪚 Open Halved Joint

Exploded and assembled views of a typical carpentry joint.

📏 Vernier Caliper Reading

Understanding the Least Count and taking measurements.

🔥 Welding Joints

Butt, Lap, T-Joint, and Corner joint configurations.

📐 Sheet Metal Funnel

3D view and 2D flat pattern development of a funnel.

💡 Staircase Wiring

Two-way switch connections for controlling a lamp from two locations.

Student Activities for Self-Learning

These activities are designed to be performed by students during self-learning hours (1.5 hours per week) to reinforce the concepts learned in the lab.

Week 1-2: Safety Familiarization +

Study: Review the safety protocols, fire extinguisher types, and PPE usage covered in Module I.
Record Work: Write a short note on the importance of safety and draw a labeled diagram of a fire extinguisher and its parts.

Week 3-4: Carpentry Tools & Alternate Materials +

Study: Identify the tools used in carpentry. Study the properties of MDF, WPC, and Plywood.
Record Work: List the tools with their uses. Prepare a table comparing MDF, WPC, and Plywood based on properties and applications.

Week 5-7: Carpentry Joint Study & Practice +

Study: Learn the step-by-step procedure for making the Open Halved Joint.
Record Work: Draw a neat sketch of the Open Halved Joint and write the complete procedure. Perform the fabrication in the lab.

Week 8-10: Fitting Shop Tools & Machines +

Study: Study machines used in the fitting shop (bench grinder, drilling machine). Study instruments (try square, surface gauge).
Record Work: List machines and power tools with their uses. Write safety precautions for any two machines.

Week 11: Fitting Joints +

Study: Study common fitting joints (L-fitting, Square gap fitting, V-joint).
Record Work: Draw neat sketches of any two fitting joints. Write the procedure for making one fitting joint.

Week 12: Revision (Fitting) +

Study: Revise tools, machines, instruments, and joint making procedures.
Record Work: Complete all drawings and notes. Write a conclusion of learning outcomes.

Week 13-15: Welding Safety & Tools

+

Study: Understand the safety practices in welding shops. Identify basic welding tools (steel rule, scriber, hammer, wire brush, welding machine).
Record Work: Write short notes on tools. Draw labeled sketches of at least 4 tools.

Week 16-18: Welding Joints & Fabrication

+

Study: Study common welding joints (Butt, Lap, Corner, T-joint). Understand the step-by-step procedure for making a butt joint.
Record Work: Draw sketches of all four welding joints. Write the procedure for making a square butt joint.

Week 19-21: Sheet Metal Tools & Operations

+

Study: Understand safety rules in the sheet metal shop. Identify tools (snips, mallet, folding bar).
Record Work: List the tools with their uses. Describe sheet metal operations like cutting, bending, and edging.

Week 22-24: Sheet Metal Fabrication

+

Study: Pattern development for a tray/box or funnel.
Record Work: Draw the development diagram of a tray and a funnel. Write the step-by-step fabrication procedure.

Week 25-27: Electrical Safety & Wiring Materials

+

Study: Understand the safety protocols during electrical work. Identify instruments used with safety protocols (testers, multimeters).
Record Work: Write safety precautions for electrical work. List the wiring materials used.

Week 28: Motor Connection

+

Study: Understand the circuit diagram and procedure for connecting a motor with a DOL starter.
Record Work: Draw the circuit diagram and write the step-by-step procedure.

Week 29-30: Electronics & Soldering +

Study: Identify electronic components (Passive and Active). Familiarise with rectifiers and filter circuits.
Record Work: Write notes on components and rectifiers. Write the procedure for soldering practice.

Total Self-Learning Hours: 45 hours (1.5 hrs × 30 weeks).

Assessment Methodology & Examination Pattern

CIE and ESE Breakdown

Mode	Component	Marks	Converted to	Schedule
CIE	Continuous Evaluation (Attendance, Timely submission, Fit & Finish, Model submission)	50	30	Every Lab
	Practical Test 1 (First half)	40	15	7th Week
	Practical Test 2 (Second half)	40	15	14th Week
ESE	Certified Record + Viva	12	40	End Semester
	Procedure Writing	4		
	Work Done + Safety	24		

Practical Test Scheme of Evaluation

CIE - Continuous Evaluation (50 Marks)

- **Preparation & Punctuality:** 5
- **Documentation:** 5
- **Timely Submission:** 5
- **Fit and Finish:** 10

- **Submission of Model:** 25

ESE - Practical Test (40 Marks)

- **Record & Viva Voce:** 12
- **Write-up / Procedure:** 4
- **Work Done (Safety, Timely, Handling):** 24

Open-ended Experiment (50 Marks)

Originality

10

Plan & Execution

20 (10+10)

Analysis & Teamwork

20 (10+10)

Module-wise Questions with Answers

Module I: Industrial Safety

Q1. What are the three methods of fire extinguishing? Explain briefly. ×

Answer: The three methods are:
1. **Cooling:** Reducing the temperature below the ignition point. Water is used for Class A fires.
2. **Smothering:** Depriving the fire of oxygen. Foam, CO2, or dry powder are used for Class B and C fires.
3. **Starvation:** Removing the fuel source from the fire.
Additionally, **Chemical Inhibition** interrupts the chemical chain reaction of the fire.

Q2. Explain the P.A.S.S. technique for operating a fire extinguisher. ×

Answer: P.A.S.S. stands for:

- **P - Pull** the safety pin.
- **A - Aim** the nozzle at the base of the fire.

- **S - Squeeze** the handle to release the agent.
- **S - Sweep** the nozzle from side to side across the base.

Q3. What is the role of PPE in a workshop?

Answer: PPE (Personal Protective Equipment) acts as the last line of defense against hazards. Examples include: Safety helmets to protect the head, goggles for eyes, gloves for hands, and safety shoes for feet. They reduce the risk of severe injuries in case of an accident.

Module II: Carpentry

Q1. What is the difference between a Rip Saw and a Cross-cut Saw?

Answer: A **Rip Saw** is designed to cut wood along the grain (lengthwise). It has larger teeth with a chisel-like cutting edge. A **Cross-cut Saw** is designed to cut wood across the grain. Its teeth are shaped like knife edges to sever the wood fibers.

Q2. Describe the properties and uses of MDF and WPC.

Answer:
MDF (Medium Density Fibreboard): Made from wood fibers and resin. It has a smooth surface, is easy to machine, and is used for furniture and cabinets. It is not water-resistant.
WPC (Wood Plastic Composite): Made from recycled plastic and wood fibers. It is highly durable, water-resistant, and eco-friendly. Used for outdoor furniture, decking, and fencing.

Q3. Write the procedure for making an Open Halved Joint.

Answer: 1. **Marking:** Mark the half-thickness and the shoulder line on both pieces. 2. **Sawing:** Cut along the marked lines using a tenon saw. 3. **Chiseling:** Remove the waste wood using a mallet and chisel. 4. **Fitting:** Test the joint and sand for a perfect fit. 5. **Assembly:** Apply glue and clamp the pieces together.

Module III: Fitting

Q1. What is the Least Count of a Vernier Caliper if 1 MSD = 1 mm and 10 divisions on the Vernier scale?

Answer: L.C. = 1 MSD - 1 VSD. Since 10 VSD = 9 MSD, 1 VSD = 0.9 mm. Therefore, L.C. = 1 mm - 0.9 mm = **0.1 mm**.

Q2. Explain the use of a Surface Plate and a Try Square in the fitting shop.

×

Answer: A **Surface Plate** provides a perfectly flat reference plane for marking and checking flatness of workpieces. A **Try Square** is used to mark and check 90-degree right angles on the workpiece.

Q3. How do you measure the thickness of a thin sheet using a Micrometer?

×

Answer: 1. Place the sheet between the anvil and the spindle. 2. Rotate the thimble until it gently grips the sheet (use the ratchet to apply consistent pressure). 3. Read the main scale on the sleeve. 4. Read the thimble scale at the horizontal line on the sleeve. 5. Add the readings (Main scale + Thimble × L.C.) to get the final dimension.

Module IV: Welding

Q1. What are the safety precautions to be taken before starting arc welding?

×

Answer: 1. Wear a welding helmet with a dark filter glass (Shade 10-13). 2. Wear leather gloves and an apron. 3. Ensure the work area is free from flammable materials. 4. Check the insulation of cables. 5. Ensure adequate ventilation to avoid inhaling fumes.

Q2. Explain the terms 'Tack Welding' and 'Square Butt Joint'.

×

Answer: Tack Welding is the process of making small, temporary weld points to hold the workpieces in alignment before the final continuous welding. **Square Butt Joint** is a joint where two metal plates are placed edge-to-edge on the same plane and welded along the seam.

Module V: Sheet Metal

Q1. Define 'Development' in sheet metal work.

×

Answer: 'Development' or 'Pattern Making' is the process of laying out the 2D flat surface of a 3D object on a sheet metal. By drawing the development accurately, the sheet metal can be cut and folded to create the desired 3D shape.

Q2. What is the difference between a Lap Seam and a Grooved Seam? ×

Answer: A **Lap Seam** is formed by overlapping the edges of two sheets and joining them together (usually by soldering or riveting). A **Grooved Seam** is made by folding the edges of two sheets together and then hammering them flat into a groove. The grooved seam is mechanically stronger and often used for tin cans.

Module VI: Electrical & Electronics

Q1. Explain the working of a stair case wiring circuit. ×

Answer: Staircase wiring uses two **two-way switches** connected together by three traveler wires. The live wire is connected to the common terminal of the first switch. The lamp is connected between the common terminal of the second switch and the neutral wire. Flipping either switch will change the state of the circuit (Open/Close), allowing the lamp to be controlled from two different locations.

Q2. What are the main components of a DOL starter? ×

Answer: The main components are: 1. **Contactor:** An electromagnetic switch that connects the motor to the supply. 2. **NO/NC Push Buttons:** For starting (NO) and stopping (NC) the motor. 3. **Overload Relay:** A protective device that trips the circuit if the motor draws excessive current.

Q3. Distinguish between Active and Passive electronic components. ×

Answer: Active components can control the flow of electricity and can amplify signals (e.g., Transistors, ICs, Diodes). **Passive components** cannot amplify signals; they only control or store energy (e.g., Resistors, Capacitors, Inductors).

Practice Model Question Paper

Based on the official examination pattern, not an official SITTTR model question paper.

Course: Engineering Workshop Practice (2009A)

Max Marks: 40 | Time: 2 Hours

Instructions to Candidates:

1. This practical examination is based on the syllabus covered in Modules I to VI.
2. Proper safety protocols and PPE usage will be evaluated as part of the "Work Done" marks.
3. Prepare the schematic diagram and step-by-step procedure on the provided blank sheet before starting the work.
4. Submit your certified record book at the end of the examination for viva voce.

Part A: Write-up / Procedure (4 Marks)

Q1. Write the aim, circuit diagram, and step-by-step procedure for **Staircase Wiring** (One lamp controlled by two switches).

Part B: Practical Work (24 Marks)

Q2. Perform the following exercise as per the given instructions. (Any one will be given by the examiner)

- **Option 1 (Carpentry):** Make an Open Halved Joint using a wooden reaper. Check the fit and finish.
- **Option 2 (Fitting):** Make a Square Fitting joint from a mild steel bar using filing and sawing operations. Ensure perfect squareness.
- **Option 3 (Welding):** Fabricate a rectangular frame using square pipes. Perform the tack welding and continuous welding operations.
- **Option 4 (Sheet Metal):** Fabricate a dust pan or tray from sheet metal. Ensure proper edging and seaming.
- **Option 5 (Electrical):** Connect a Single Phase Induction Motor to a DOL Starter. Test the Start/Stop control.

Part C: Viva Voce & Record (12 Marks)

- **Record:** Complete and neat submission of all experiments with proper certification.
- **Viva Voce:** Oral questions on the tools, safety, and working principles related to the experiment performed.

Quick Revision Cards

Module I: Safety

- **P.A.S.S.** - Pull, Aim, Squeeze, Sweep.
- **Fire Classes:** A (Wood), B (Liquids), C (Gases), D (Metals), K (Kitchen).
- Always wear PPE (Helmet, Goggles, Gloves, Shoes).

CO1

Module II: Carpentry

- **Alternate Materials:** MDF (Resin + Fiber), WPC (Plastic + Wood).
- **Joint:** Open Halved Joint - interlocking half-thickness cuts.
- **Tools:** Jack plane (smoothing), Chisel (carving), Tenon saw (fine cuts).

CO1

Module III: Fitting

- **Precision Tools:** Vernier Caliper (L.C. 0.02mm), Micrometer (L.C. 0.01mm).
- **Operations:** Marking, Cutting (Hacksaw), Filing (Flat/Round files).
- Always check squareness with a Try Square.

CO2

Module IV: Welding

- **Safety:** Never look at the arc without a helmet.
- **Joints:** Butt, Lap, T, Corner.
- **Operation:** Strike arc, maintain constant speed, clean slag with chipping hammer.

CO3

Module V: Sheet Metal

- **Development:** Unfolding 3D shapes onto flat sheet metal.
- **Tools:** Snips (cutting), Mallet (forming), Stakes (support).
- **Models:** Tray, Box, Funnel - uses bending, edging, and seaming.

CO4

Module VI: Electrical & Electronics

- **Wiring:** Staircase wiring uses 2-way switches.
- **Motor:** DOL starter uses Contactor + NO/NC + Overload Relay.
- **Soldering:** Heat joint → Apply solder → Let cool (shiny joint).

CO5

Common Mistakes to Avoid

- **Safety:** Forgetting to wear safety goggles or using the wrong fire extinguisher for the fire class.
- **Carpentry:** Cutting the wood too short or sawing outside the marked line.
- **Fitting:** Using a coarse file on a finished surface, ruining the precision.
- **Welding:** Moving the electrode too fast or too slow, leading to a weak or porous weld bead.
- **Sheet Metal:** Not adding bend allowances to the development, resulting in a smaller final product.
- **Electrical:** Not tightening the screw terminals properly, causing arcing and overheating.

Glossary of Important Terms

Term	Meaning
Arc Welding	A welding process that uses an electric arc to join metal parts.
CIE	Continuous Internal Evaluation - Ongoing assessment during the semester.
DOL Starter	Direct On Line starter - connects an induction motor directly to the power supply.
ESE	End Semester Examination - Final practical examination.
Least Count (L.C.)	The smallest value that can be measured by a precision instrument.
MDF	Medium Density Fibreboard - An engineered wood product.
PPE	Personal Protective Equipment - Gear worn to minimize exposure to hazards.
Rectifier	An electronic circuit that converts AC current to DC current.
Slag	The solidified residue left on top of a weld bead, which needs to be chipped off.
Soldering	A process of joining two metals by melting a filler metal (solder) into the joint.
Tack Weld	A small, temporary weld used to hold parts together before final welding.
Vernier Caliper	A precision measuring tool used to measure internal and external dimensions.
WPC	Wood Plastic Composite - A composite material made from wood fiber and plastic.

Learning Resources

Textbooks

- Hajara Chaudhary, S.K. - *Workshop Technology* (Median Promoters and publishers, New Delhi, 2015).
- Gupta, J.B. - *Electrical Installation Estimation and Costing* (S.K. Kataria & Sons).

Reference Books

- Raghuwanshi, B.S. - *Workshop Technology* (Dhanpat Rai and sons, New Delhi, 2014).
- Reddy, K. Venkat - *Work Practice Manual* (BS Publications, Hyderabad, 2014).

Web Resources

- Bharat Skills Study Material - https://bharatskills.gov.in (Modules I to VI).